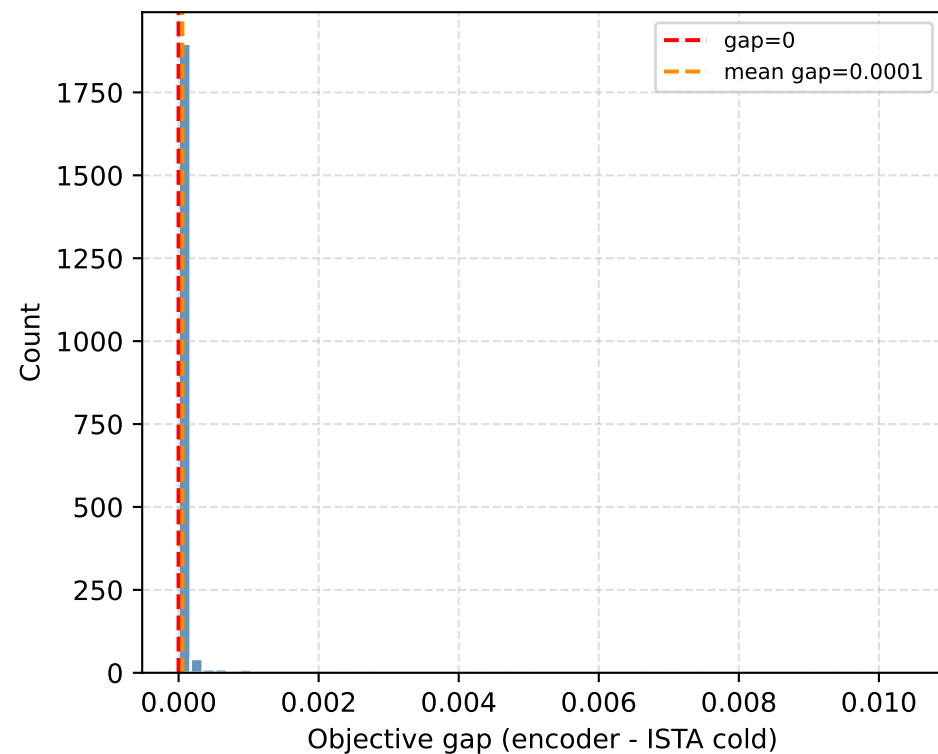
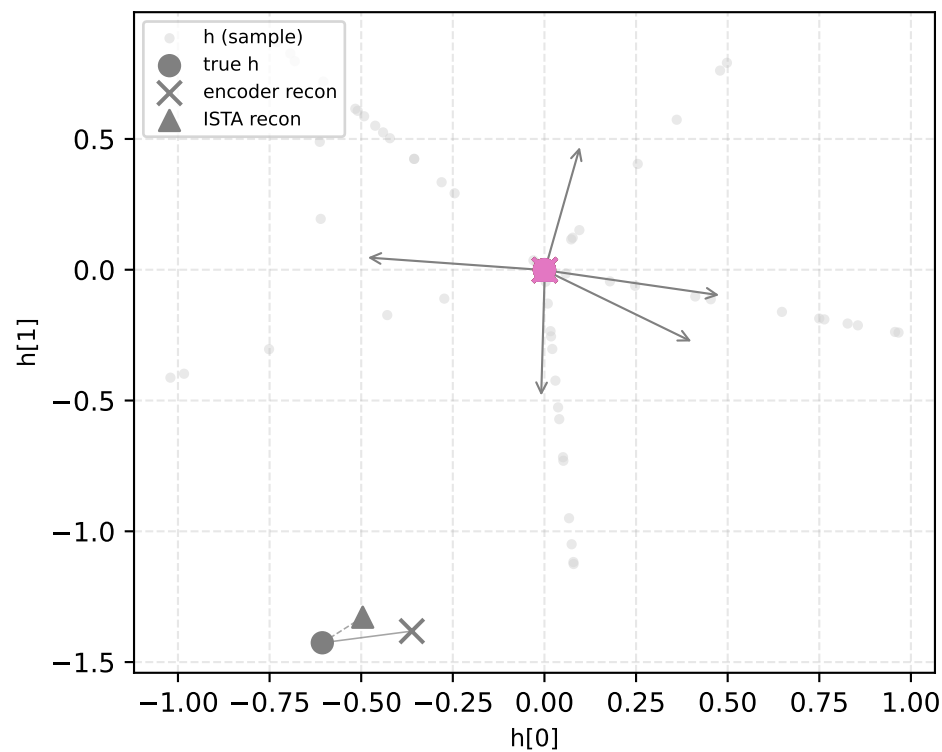


Amortization Gap — SAE Encoder vs Optimal ISTA Code (lambda=0.1)  
Toy superposition: 5 features, 2-D hidden

(a) Per-sample amortization gap  
(encoder - ISTA cold)



(b) 2-D geometry: encoder vs ISTA  
(circle=h, x=enc recon, tri=ISTA cold recon)



(c) Per-sample objectives  
(encoder always  $\geq$  ISTA)

